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Prevalence and correlates of diabetes distress and depressive symptoms among individuals with type-2 diabetes mellitus during Ramadan fasting: A cross-sectional study in Bangladesh amid the COVID-19

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A R T I C L E I N F O

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A B S T R A C T

Aims: Psychological concerns relating to “diabetes distress” (DD) and depressive symptoms (DS) in individuals with type-2 diabetes mellitus (T2DM) may negatively impact adherence to medical treatments and overall mental health. Thus, this study was undertaken to investigate DS and DD in relation to fasting during the month of Ramadan.

Methods: A cross-sectional survey was conducted among 735 patients with T2DM in 2021. DD and DS were measured by the Problem Areas in Diabetes scale and Patient Health Questionnaire-9, respectively. Logistic regression and correlation analyses were executed.

Results: More than one-third of the participants (41.2%) had DD and DS (36.9%). DS was significantly higher in participants who did not fast ($p = 0.027$). Participants who had higher dietary diversity were less likely to have DD ($p = 0.004$) and DS ($p = 0.001$). Females (AOR = 1.89, 95% CI: 1.25–2.85) and those who lived alone (AOR = 1.89, 95% CI: 1.25–2.85) were more likely to have DS. Participants with diabetes-related complications were more likely to experience DS (AOR = 2.17; 95% CI: 1.5–3.13) and DD (AOR = 3.46; 95% CI: 2.42–4.95). DD was also associated with being younger ($p = 0.003$), having hypertension ($p = 0.030$), having heart disease ($p = 0.012$), and taking insulin ($p = 0.010$).

Conclusions: Individuals with T2DM who were not fasting experienced more mental health concerns. Psychosocial support and other interventions from health professionals should be examined and empirical interventions should be implemented to promote the mental health and well-being of individuals with T2DM.

1. Background

Type 2 diabetes mellitus (T2DM) is a chronic and complex metabolic condition. T2DM prevalence estimates have risen to epidemic levels in recent decades, including in developing and newly industrialized countries like Bangladesh [1]. According to the International Diabetes Federation (IDF), diabetes currently affects 382 million people globally, which is anticipated to rise 592 million by 2035 [2] and >80% of diabetes deaths occurring in low- and middle-income countries (LMICs). Recent estimates include 7.1 million individuals with diabetes in

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Bangladesh, with 3.7 million undiagnosed cases and approximately 129,000 fatalities [3]. Existing data suggest that common mental health problems are frequently experienced by people with T2DM, with depressive symptoms (DS) reported by 17.6% of people with T2DM as compared to 9.8% in those without [4]. A systematic review that included LMICs like India, Mexico, Brazil and China found that depression among people with T2DM was greater in LMICs than in higher-income countries [5]. In South Asia, the prevalence of depression has been reported to the range from 5.9% to 49% [6–10]. Depression in people with T2DM has been linked to hyperglycemia and diabetes-related complications such as retinopathy, nephropathy, neuropathy, and macrovascular difficulties [11], as well as premature death [12]. A prior study also indicated that individuals with T2DM experience high levels of diabetes distress (DD) [13]. DD involves patients' concerns about disease management, support, emotional load, and access to care and differs from depression [14]. When personal efforts to mitigate these challenges fail to succeed as expected, or when diabetic complications exert tolls on physical and mental health, the well-being and quality of life of individuals with T2DM diabetes patients may be adversely impacted [15].

In addition, evidence showed that mental health difficulties may be particularly high in people with T2DM during the COVID-19 pandemic due to an elevated risk of mortality and morbidity from COVID-19 infection [16]. To date, the COVID-19 pandemic has severely impacted Bangladesh with 773,513 confirmed cases and 11,934 deaths as of May 9, 2021 [17]. During the pandemic, Bangladesh has implemented social isolation and home quarantine measures, which has limited normal physical movement and healthcare access [18]. Thus, many people with T2DM have experienced more difficulties in regulating glycemic levels, perhaps due to more limited access to hospitals during the pandemic [19].

Though there is a paucity of research about DS among patients with T2DM during the pandemic in Bangladesh, some previous studies from South Asian settings before the COVID-19 pandemic suggest that the prevalence of depression in people with diabetes was two or more times higher than in people without diabetes [8,20]. A Bangladeshi study conducted in 2011 involving individuals with T2DM found that 34–36% experienced DS, with correlates of DS including age, income, gender, treatment intensity, and co-occurring disorders [21]. Despite these mental health concerns, most diabetes care guidelines focus on medical aspects of primary management without considering psychological functioning [22].

Fasting during the month of Ramadan has been found to be beneficial both to physical health [23] and mental and social well-being [24]. Ramadan is a holy month for Muslims, and fasting throughout Ramadan is one of the five fundamental pillars of Islam [25]. During this holy month, Muslims fast from dawn to sunset for 30 consecutive days annually, and during fasting, people with T2DM typically adhere better to their recommended diabetic care plans [26,27]. Previous research showed that fasting during month of Ramadan was significantly associated with reduced depression and DD in individuals with T2DM [27]. These findings have been interpreted as suggesting that aspects of spirituality, self-discipline, self-control and disease management may help with addressing depression and improving diabetic control [27]. As diabetes and depression are both chronic conditions, researchers are investigating empirically supported strategies for preventing the co-occurrence of and interactions between depression and diabetes [28,29].

Past research revealed that spirituality like fasting during the month of Ramadan can be implemented to prevent depression and DD [27], improve glycemic control [30] and further diabetic management in T2DM patients. Several studies have been conducted exploring potential positive impacts of Ramadan fasting on mental health among individuals with T2DM in countries like Pakistan [31] and Kuwait [27]. However, studies among individuals with T2DM in Bangladesh during the month of Ramadan are limited. Such information could assist in planning and adopting appropriate interventions to mitigate the negative psychological impacts arising from DS and DD in people with T2DM in a poor resource country setting like Bangladesh.

Hence, we performed this study during the month of Ramadan to determine the prevalence and identify factors associated with DS and DD amongst individuals with T2DM. We also aimed to examine how fasting related to DS and DD in individuals with T2DM during Ramadan.

2. Methods

2.1. Study design and participants

A cross-sectional study was conducted with a convenience sampling technique. The patients with T2DM who were physically and mentally capable of fasting during the month of Ramadan, aged between 18 and 65 years, able to read and understand Bengali, and living in Bangladesh during the survey were eligible for the study. Patients with T2DM across Bangladesh participated. All study procedures adhered to the ethical requirements of the applicable national and institutional committees on human experimentation, as well as the Declaration of Helsinki. The study protocol was reviewed and approved by the Biosafety, Biosecurity & Ethical Committee, Jahangirnagar University, Dhaka, Bangladesh [Ref No: BBEC, JU/M 2021/3(2)].

2.2. Sampling

As we had two primary outcomes (DS and DD), we calculated two different sample sizes for the two different outcomes. For DS, a 34% prevalence from a study among individuals with T2DM [21], 95% confidence interval, and 4% margin of error were used to calculate the needed sample size as 539. For DD, a 48.5% prevalence among individuals with T2DM [32], 95% confidence interval, and 4% margin of error were used to calculate the needed sample size as 600. A total of 749 individuals with T2DM participated and data from 735 were included in final analyses after removing incomplete or ineligible data.

2.3. Procedure

Data were collected through a telephone survey during the month of Ramadan between May 5 and May 13, 2021 using a semi-structured questionnaire. An orientation training was conducted with 37 research assistants by the principal investigator on survey methods and data collection techniques prior to data collection. The questionnaire consisted of socio-demographic information, additional information regarding Ramadan and fasting, a questionnaire based on FAO 2013 guidelines [39] to measure dietary diversity, the Patient Health Questionnaire-9 [44,45], and the Problem Areas in Diabetes questionnaire [41]. Initially, the questionnaire was developed in English from previous literature, then translated into the local language (Bengali). It was translated by a bilingual member of the research team and it was then double-checked by another bilingual team member. A third independent bilingual translator back-translated the material to ensure consistency and obviate biases. The Bangla version of the questionnaire was used in the study. Finally, the questionnaire was pretested with a small sample of participants (20 participants) to confirm its clarity and utility. Minor modifications were incorporated in the questionnaire after pretesting. Pretesting data were excluded from the final analysis. After the finalization of the questionnaire, the research assistants conducted the telephone surveys with their relatives and acquaintances who had T2DM. The participants were asked if they had T2DM based on physicians' clinical diagnosis as well as the World Health Organization (WHO) clinical criteria [33]. Participants who did not have clinical diagnoses and did not adhere to the WHO clinical criteria on T2DM were excluded from this study. After a brief explanation about the study aims and procedure, informed verbal consent was sought from each individual prior to data collection. Participants were informed that anonymity and confidentiality would be maintained and data were kept in a password-protected folder for ensuring security. Study participation was voluntary. Data were checked and rechecked at the end of each day of collecting data to ensure that questionnaires were filled completely and consistently.

2.4. Measures

2.4.1. Socio-demographic information

Socio-demographic information included age (<50 years, 50–65 years, and > 65 years) [34], sex (female and male), educational status (no formal education, up to secondary, up to higher secondary, Honors, Masters or above) [35], monthly family income (<15,000 BDT, 15,000–30,000 BDT, and > 30,000 BDT) [36], living status (living with spouse/ children and alone). Moreover, self-reported information about co-occurring disorders (e.g., hypertension, heart disease, stroke, kidney disease) was acquired from participants.

2.4.2. Diabetes related information and additional variables

Participants were asked about their duration of having diabetes (≤1 year, 1.1–5 years, 5.1–10 years, ≥ 10.1 years), and presence of diabetes-related complications (retinopathy, nephropathy, neuropathy, macrovascular complications, and sexual dysfunction) with dichotomous responses (yes/no). Participants were asked about their types of management plans: i) diet, ii) lifestyle modifications (including physical exercising, dieting) iii) oral medications, and iv.) insulin, were assessed through a checklist where multiple responses could be chosen. Self-reported measures were used to assess participants' height and weight [37]. Body mass index (BMI) was determined as weight (kg) divided by the square of height in meters and classified as underweight (<BMI 18.5 kg/m^2), normal weight (BMI = 18.5–24.9 kg/m^2), overweight (BMI = 25.0–29.9 kg/m^2), and obese (BMI ≥ 30 kg/m^2) [38]. Additional information regarding Ramadan and fasting was collected and included whether yes/no responses regarding whether COVID-19 influenced their decisions about their fasting, they were unable despite wanting to fast because of diabetes-related complications, and they prayed Taraweeh during Ramadan. Participants were also asked how they perceived their weight compared to the pre-COVID-19 period with four possible responses (unaware/increased/decreased/same).

2.4.3. Dietary diversity

To measure dietary diversity, a questionnaire was used following FAO (2013) guidelines [39]. Nine major food groups were included in the questionnaire: (i) cereals, (ii) dark green, leafy vegetables; (iii) vitamin-A-rich fruit and vegetables; (iv) other fruit and vegetables; (v) organ meat; (vi) fish and seafood; (viii) eggs; (viii) legumes, nuts and grains; and (ix) milk and milk products. Based on the food groups consumed in the last 24 h, the dietary diversity score (DDS) was estimated based on a half-serving of at least one item from each of the food groups indicated. Each food group consumed was given a one-to-nine score. The total of all food groups was the DDS [40].

2.4.4. Problem Areas in diabetes Scale—Five-item short form (PAID-5)

The PAID-5 questionnaire was used to assess DD [41]. It is a reliable and valid shortened version of the original 20-item PAID scale [42], containing five of the emotional-distress items from the PAID scale. Psychometric properties of the scale were originally assessed in the Diabetes Attitudes Wishes and Needs Monitoring Individual Needs in Diabetes (DAWN MIND) study [43]. The PAID-5 has a five-point response option (0–4 representing 'Not a problem' to 'A serious problem') with a total score range of 0 to 20, where higher scores indicate more diabetes-related emotional distress [41]. A cutoff of ≥ 8 was used to indicate the presence of DD [41]. This cutoff score has demonstrated 95% sensitivity and 89% specificity [41]. In the present study, the Cronbach's alpha of the scale was 0.896.

2.4.5. Patient health questionnaire (PHQ-9)

A Bangla validated nine-item PHQ-9 scale was used to assess DS [44,45]. The PHQ-9 is based on the Diagnostic and Statistical Manual of Mental Disorders, 4th Edition (DSM-IV) diagnostic criteria for depression. The scale has been demonstrated to have high sensitivity and specificity in identifying cases of depression [46]. Respondents answered on a four-point Likert scale ranging from "0 = not at all" to "3 = virtually every day" with a focus on the past two weeks. The total score varies between 0 and 27. The following five cut-off points were used to categorize DS: i) '0–4′ for 'normal'; ii) '5–9′ for 'mild DS'; iii) '10–14′ for 'moderate DS'; iv) '15–19′ for 'moderately severe DS'; and finally, v) '20 or higher' for 'severe DS′. A cutoff score ≥ 10 was used to record DS in participants [47]. This scale was previously used among T2DM patients in Bangladesh [21] and during the month of Ramadan [27]. In the present study, the Cronbach's alpha of the scale was 0.82.

2.5. Statistical analysis

All analyses were executed using Statistical Package for the Social Sciences (SPSS version 25). Descriptive statistics were used to calculate the frequency of sample characteristics of the study population. Chi-square and reliability tests (Cronbach's alpha) were executed as first-order analyses. The chi-square test was used to assess differences in characteristics between groups. Multivariable logistic regression analysis was performed to statistically predict the odds of DD and DS (outcome variables) with a 95% confidence interval (CI). Dependent variables were binary in nature and multicollinearity was checked before proceeding with analysis. Correlation analyses were conducted to determine relationships between DS and DD. A p-value < 0.05 was considered significant given the exploratory nature of the study.

3. Results

3.1. General characteristics

A total of 735 subjects aged 18–89 years (mean age = 52.30 years [SD = 12.02]) were included in the final analyses. Most were female (54.3%) and married (84.8%) (Table 1). Most reported having hypertension (57.1%). Sizable minorities reported normal BMI (46.8%) and perceived decreased weight compared to pre-Ramadan period (33.2%). Most used oral medications for diabetes control (77%) and reported fasting regularly during Ramadan (89.3%) (Table 1).

3.2. Diabetes distress and depressive symptoms

DD and DS were reported by 41.2% and 36.9%, respectively. Severities of depression were reported by the following percentages of respondents: mild (34%), moderate (21.1%), moderately severe (11.8%), and severe (3.9%). More than one-fourth of participants (26.3%) had both DD and DS, whereas 10.6% had only DD and 15.0% had only DS (Table 1). Sex, education, living status, co-occurring disorders, treatments for diabetes control, having diabetes-related complications and fasting variables were significantly associated with DD and DS (Table 1). DD and DS were positively correlated with each other ($r = 0.593$; $p < 0.001$) (Fig. 1).

3.3. Factors associated with diabetes distress

Younger participants (aged < 50 years, 50–65 years) had greater odds of DD compared to older participants (>65 years) (AOR: 2.94; 95% CI: 1.45–5.97, $p = 0.003$; AOR: 2.46; 95% CI: 1.32–4.59, $p = 0.004$). Participants who lived alone were more likely to have DD compared to those who resided with spouse/children (AOR: 2.04; 95% CI: 1.01–4.13, $p = 0.046$). In terms of co-occurring conditions, those with heart disease had greater odds of DD compared to those without (AOR: 1.66; 95% CI: 1.12–2.50, $p = 0.012$). Similar findings were observed with respect to

Table 1
Characteristics of participants, overall and by diabetes distress (DD) and depressive symptoms (DS).

Variables	Overall N = 735	DS & DD absent n = 354; 48.2%	Only DS present n = 110; 15.0%	Only DD present n = 78; 10.6%	Both DS & DD present N = 193; 26.3%	*p*-value
Age						
< 50 years	281 (38.2)	131 (37)	39 (35.5)	29 (37.2)	82 (42.5)	0.235
50–65 years	371 (50.5)	180 (50.8)	65 (59.1)	38 (48.7)	88 (45.6)	
> 65 years	83 (11.3)	43 (12.1)	6 (5.5)	11 (14.1)	23 (11.9)	
Sex						
Male	336 (45.7)	178 (50.3)	50 (45.5)	37 (47.4)	71 (36.8)	**0.026**
Female	399 (54.3)	176 (49.7)	60 (54.5)	41 (52.6)	122 (63.2)	
Marital status						
Married	623 (84.8)	306 (86.4)	94 (85.5)	66 (84.6)	157 (81.3)	0.132
Unmarried	28 (3.8)	14 (4)	0 (0)	5 (6.4)	9 (4.7)	
Divorced/widowed	84 (11.4)	34 (9.6)	16 (14.5)	7 (9)	27 (14)	
Education level						
No formal education	45 (6.1)	18 (5.1)	5 (4.5)	7 (9)	15 (7.8)	**0.031**
Up to secondary	309 (42.0)	132 (37.3)	51 (46.4)	32 (41)	94 (48.7)	
Up to intermediate	129 (17.6)	63 (17.8)	17 (15.5)	12 (15.4)	37 (19.2)	
Honors	136 (18.5)	67 (18.9)	21	17 (21.8)	31 (16.1)	
Masters or above	116 (15.8)	74 (20.9)	16 (14.5)	10 (12.8)	16 (8.3)	
Family income						
<15,000 BDT	108 (14.7)	45 (12.7)	13 (11.8)	15 (19.2)	35 (18.1)	0.072
15,000–30,000 BDT	132 (18)	54 (15.3)	20 (18.2)	19 (24.4)	39 (20.2)	
>30,000 BDT	495 (67.3)	255 (72)	77 (70)	44 (56.4)	119 (61.7)	
Living status						
Living with spouse/ children	681 (92.7)	334 (94.4)	108 (98.2)	72 (92.3)	167 (86.5)	**0.001**
Alone	54 (7.3)	20 (5.6)	2 (1.8)	6 (7.7)	26 (13.5)	
Co-occurring disorders						
Hypertension	420 (57.1)	167 (47.2)	62 (56.4)	51 (65.4)	140 (72.5)	**<0.001**
Heart disease	182 (24.8)	58 (16.4)	30 (27.3)	23 (29.5)	71 (36.8)	**<0.001**
Stroke	74 (10.1)	21 (5.9)	12 (10.9)	15 (19.2)	26 (13.5)	**0.001**
Kidney disease	66 (9)	21 (5.9)	5 (4.5)	11 (14.1)	29 (15)	**<0.001**
Perceived weight compared to pre-Ramadan period						
Increased	86 (11.7)	45 (12.7)	8 (7.3)	10 (12.8)	23 (11.9)	0.461
Decreased	244 (33.2)	105 (29.7)	40 (36.4)	30 (38.5)	69 (35.8)	
Same	239 (32.5)	122 (34.5)	37 (33.6)	26 (33.3)	54 (28)	
Unaware	166 (22.6)	82 (23.2)	25 (22.7)	12 (15.4)	47 (24.4)	
Duration of diabetes						
≤ 1 year	86 (11.7)	50 (14.1)	9 (8.2)	10 (12.8)	17 (8.8)	0.268
1.1–5 years	277 (37.7)	138 (39)	47 (42.7)	23 (29.5)	69 (35.8)	
5.1–10 years	195 (26.5)	90 (25.4)	28 (25.5)	20 (25.6)	57 (29.5)	
≥ 10.1 years	177 (24.1)	76 (21.5)	26 (23.6)	25 (32.1)	50 (25.9)	
Treatment for diabetes control						
Diet	528 (71.8)	250 (70.6)	80 (72.7)	56 (71.8)	142 (73.6)	0.899
Lifestyle modification	546 (74.3)	262 (74)	81 (73.6)	59 (75.6)	144 (74.6)	0.989
Oral medications	566 (77)	261 (73.7)	97 (88.2)	60 (76.9)	148 (76.7)	**0.019**
Insulin	235 (32)	84 (23.7)	35 (31.8)	26 (33.3)	90 (46.6)	**<0.001**
BMI						
Normal	344 (46.8)	167 (47.2)	55 (50)	32 (41)	90 (46.6)	0.068
Underweight	18 (2.4)	5 (1.4)	0 (0)	6 (7.7)	7 (3.6)	
Overweight	278 (37.8)	134 (37.9)	39 (35.5)	31 (39.7)	74 (38.3)	
Obese	95 (12.9)	48 (13.6)	16 (14.5)	9 (11.5)	22 (11.4)	
Having diabetes-related complications[€]						
Yes	303 (41.2)	86 (24.3)	61 (55.5)	35 (44.9)	121 (62.7)	**<0.001**
No	432 (58.8)	268 (75.7)	49 (44.5)	43 (55.1)	72 (37.3)	
Fasting						
Yes	656 (89.3)	322 (91.0)	102 (92.7)	70 (89.7)	162 (83.9)	**0.043**
No	79 (10.7)	32 (9.0)	8 (7.3)	8 (10.3)	31 (16.1)	

[€] Diabetes related complications includes retinopathy, nephropathy, neuropathy, macrovascular difficulties.

hypertension (AOR: 1.49; 95% CI: 1.04–2.14, p = 0.030). Likewise, participants with diabetes-related complications were more likely than those without to have DD (AOR: 3.46; 95% CI: 2.42–4.95, $p < 0.001$). In addition, those who took insulin for diabetes control were more likely than those not taking insulin to have DD (AOR: 1.69; 95% CI: 1.13–2.52, p = 0.010) (Table 2). DD was negatively associated with dietary diversity (AOR: 0.87; 95% CI: 0.79–0.96, p = 0.004) (Table 4).

3.4. Factors associated with depressive symptoms

Females were more likely than males to have DS (AOR: 1.89; 95% CI: 1.25–2.85, p = 0.002). Participants with Masters or higher education were less likely to have DS than those who had no formal education (AOR: 0.32; 95% CI: 0.13–0.74, p = 0.008). Participants who lived alone were more likely to have DS compared to those who resided with spouse/children (AOR: 2.6; 95% CI: 1.27–5.35, p = 0.009). Those with hypertension compared to those without had greater odds of DS (AOR: 1.9; 95% CI: 1.31–2.76, p = 0.001). Similar findings were observed with respect to heart disease (AOR: 1.67; 95% CI: 1.11–2.50, p = 0.014). Participants with diabetes-related complications compared to those without were more likely to have DS (AOR: 2.17; 95% CI: 1.51–3.13, $p <$ 0.001). Those who reported underweight than who reported normal weight were more likely to have DS (AOR: 4.12; 95% CI: 1.20–14.13, p = 0.025) (Table 3). DS were negatively associated with dietary diversity (AOR: 0.84; 95% CI: 0.76–0.93, p = 0.001). Participants who did not fast during the month of Ramadan were more likely to have DS compared to those who fasted (AOR: 1.71; 95% CI: 1.06–2.75, p = 0.027) (Table 4).

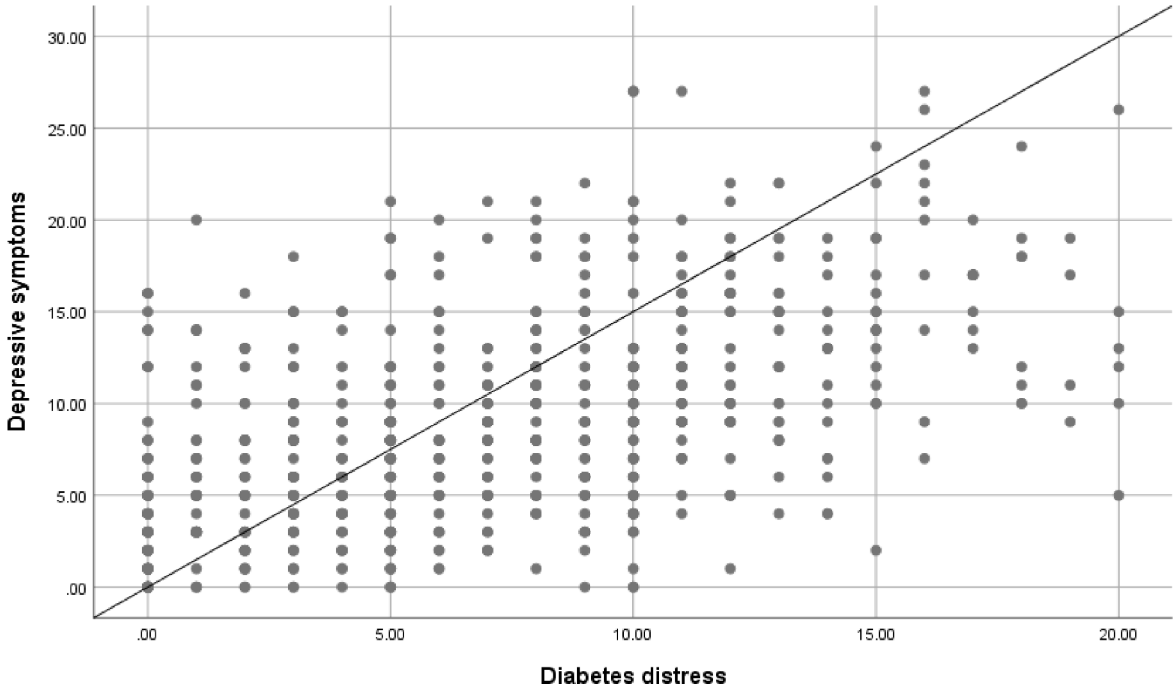

Fig. 1. Scatter diagram showing correlation between diabetes distress and depressive symptoms.

4. Discussion

DD is an emotional burden when living chronically with T2DM [48], and assessing DD and DS is important to facilitate addressing needs of individuals with T2DM [48]. The present study aimed to elucidate the prevalence and factors associated with DD and DS among patients with T2DM during the intermittent fasting of Ramadan in the setting of the COVID-19 pandemic. We found that 41.2% and 36.9% of the participants had DD and DS, respectively. While clinical depression is common among individuals with T2DM, most individuals are not clinically depressed, but rather experience less severe symptoms and are distressed about their long-standing T2DM, its management and complications [49].

The study revealed that 41.2% of the participants had DD during the month of Ramadan, approximating but slightly lower than the percentage (48.5%) found in a prior Bangladeshi study among individuals with T2DM (48.5%) [32]. Similarly, the percentage of individuals with DS observed in the present study is lower than that in a prior study among individuals with T2DM (36.9% vs. 61.9%) [50]. Speculatively, the lower percentages may reflect the present study having been undertaken during Ramadan given that fasting was negatively associated with DS in the present study. Consistent with this interpretation, a previous study during the month of Ramadan among individuals with T2DM found fasting may contribute to lowering DS (decreases in PHQ-9 score by 3.57 points) and DD (decrease in PAID score by 5.13 points) [27]. This finding suggests the importance of conducting a longitudinal study by gathering data before, during and after Ramadan fasting to understand better precise relationships between fasting, DD, DS and other factors.

Our study showed that 89.3% of the total respondents were fasting during the month of Ramadan. Of 656 respondents who reported fasting, 322 participants (49.08%) had no DS or DD, and 162 participants (24.69%) had both DD and DS, and the latter is higher than found in a previous study (5.3%) [34]. Regular screening for DD and DS should be enhanced to ensure better mental health with psychological and medical intervention as lower detection rate and under-recognition are considered to be an added barrier to diabetes management of T2DM in primary care [51]. Governments, non-government organizations and other groups could design and enact awareness programs to provide health education and improve T2DM-related health literacy. Internists and other healthcare providers could help educate newly diagnosed patients about the associated burden of DD and DS. Counselling of patient could be started at this point, as indicated.

In the present study, the percentage of having both DD and DS were significantly higher (39.24% vs. 24.69%) in participants who didn't fast which is compatible to previous studies [27,31]. An inverse relationship between depression and fasting was also observed in a previous study of nurses, suggesting that fasting may improve psychological problems such as depression [29]. Fewer studies, however, have explored relationships between fasting and DD. Some have speculated that fasting may share features with meditation-based relaxation techniques and mindfulness [27,52] which might help reduce DD and DS in participants who fasted. Another explanation may involve Ramadan being a time for social cohesion and support [53], which has been shown to be an effective non-pharmacological therapy for psychological issues like DD and DS [54]. However, future studies are needed to examine these currently speculative possibilities.

We observed that women with T2DM had nearly a two-fold increased odds of experiencing DS compared to males, consistent with previous Bangladeshi findings [21]. The increased frequency of DS among women might be affected by sociocultural, psychological, biological and hormonal factors, among other domains [10]. We also observed that younger participants were more likely to experience DD than older individuals which accords to previous findings [32]. Underweight participants were 4.11 times more likely to experience DS, which is also consistent with prior findings [55]. According to a systematic review of potential T2DM risk factors, Bangladeshi individuals may have greater T2DM risk at lower BMIs than Western adults [56]. A possible explanation behind the link between being underweight and experiencing DS is that depressed individual may consume less foods compared to other BMI groups [55].

Inverse relationships between dietary diversity and both DD and DS were found, suggesting that diets involving multiple types of food may contribute to better mental health in people with T2DM. Consistent with our findings, a prior study in a developing country demonstrated that the odds of severe depression among women were reduced by 39% for

Table 2
Factors associated (multivariable logistic regression) for diabetes distress among patients with type-2 diabetes.

Variables	AOR (95% CI)	*p*-value
Age		
> 65 years	Ref.	
< 50 years	2.941 (1.449–5.967)	**0.003**
50–65 years	2.464 (1.324–4.585)	**0.004**
Sex		
Male	Ref.	
Female	1.308 (0.881–1.94)	0.183
Marital status		
Unmarried	Ref.	
Married	1.706 (0.598–4.866)	0.318
Divorced/widowed	2.294 (0.704–7.473)	0.168
Education level		
No formal education	Ref.	
Up to secondary	1.034 (0.508–2.105)	0.927
Up to intermediate	1.01 (0.458–2.227)	0.981
Honors	0.869 (0.392–1.929)	0.731
Masters or above	0.544 (0.235–1.261)	0.156
Family income		
<15,000 BDT	Ref.	
15,000–30,000 BDT	0.866 (0.487–1.539)	0.624
>30,000 BDT	0.958 (0.583–1.573)	0.864
Living status		
Living with spouse/ children	Ref.	
Alone	2.043 (1.011–4.128)	**0.046**
Co-occurring disorders (Ref. No)		
Hypertension	1.491 (1.04–2.136)	**0.030**
Heart disease	1.675 (1.121–2.502)	**0.012**
Stroke	0.96 (0.545–1.694)	0.889
Kidney disease	0.918 (0.511–1.651)	0.776
Perceived weight compared to pre-Ramadan period		
Unaware	Ref.	
Increased	0.569 (0.304–1.066)	0.078
Decreased	0.904 (0.573–1.425)	0.663
Same	0.915 (0.579–1.445)	0.702
Duration of diabetes		
≤ 1 year	Ref.	
1.1–5 years	1.497 (0.822–2.726)	0.188
5.1–10 years	1.372 (0.722–2.608)	0.335
≥ 10.1 years	1.381 (0.69–2.763)	0.362
Treatment for diabetes control (Ref. No)		
Diet	1.19 (0.809–1.751)	0.376
Lifestyle modification	0.878 (0.592–1.303)	0.517
Oral medications	1.523 (0.993–2.334)	0.054
Insulin	1.688 (1.131–2.52)	**0.010**
BMI		
Normal	Ref.	
Underweight	0.705 (0.21–2.359)	0.570
Overweight	0.892 (0.615–1.294)	0.548
Obese	0.694 (0.407–1.182)	0.179
Having diabetes-related complications[€]		
No	Ref.	
Yes	3.459 (2.416–4.952)	**<0.001**
COVID-19 influenced decision about fasting		
No	Ref.	
Yes	1.362 (0.688–2.699)	0.376
Cannot fast even though wanted to fast because of diabetes-related complications		
No	Ref.	
Yes	1.776 (0.91–3.464)	0.092
Praying Taraweeh Prayers during this Ramadan		
No	Ref.	
Yes	1.154 (0.795–1.673)	0.451

Note: AOR = Adjusted odds ratio; CI = Confidence interval; BDT = Bangladeshi Taka.
[€] Diabetes-related complications include retinopathy, nephropathy, neuropathy, and macrovascular difficulties.

Table 3
Factors associated (multivariable logistic regression) for depressive symptoms among patients with type-2 diabetes.

Variables	AOR (95% CI)	*p*-value
Age		
> 65 years	Ref.	
< 50 years	1.598 (0.788–3.241)	0.194
50–65 years	1.296 (0.7–2.403)	0.409
Sex		
Male	Ref.	
Female	1.889 (1.252–2.85)	**0.002**
Marital status		
Unmarried	Ref.	
Married	0.493 (0.178–1.364)	0.173
Divorced/widowed	0.334 (0.104–1.073)	0.065
Education level		
No formal education	Ref.	
Up to secondary	0.574 (0.285–1.156)	0.120
Up to intermediate	0.636 (0.29–1.395)	0.259
Honors	0.456 (0.205–1.015)	0.054
Masters or above	0.315 (0.134–0.739)	**0.008**
Family income		
<15,000 BDT	Ref.	
15,000–30,000 BDT	0.761 (0.425–1.361)	0.357
>30,000 BDT	0.607 (0.366–1.006)	0.053
Living status		
Living with spouse/ children	Ref.	
Alone	2.6 (1.265–5.345)	**0.009**
Co-occurring disorders (Ref. No)		
Hypertension	1.9 (1.307–2.761)	**0.001**
Heart disease	1.666 (1.11–2.498)	**0.014**
Stroke	1.3 (0.726–2.328)	0.377
Kidney disease	1.718 (0.952–3.101)	0.072
Perceived weight compared to pre-Ramadan period		
Unaware	Ref.	
Increased	0.937 (0.494–1.775)	0.841
Decreased	1.261 (0.79–2.013)	0.331
Same	1.109 (0.691–1.779)	0.669
Duration of diabetes		
≤ 1 year	Ref.	
1.1–5 years	1.217 (0.656–2.259)	0.533
5.1–10 years	1.737 (0.901–3.349)	0.099
≥ 10.1 years	1.992 (0.988–4.018)	0.054
Treatment for diabetes control (Ref. No)		
Diet	1.099 (0.743–1.627)	0.635
Lifestyle modification	0.892 (0.596–1.335)	0.578
Oral medications	0.959 (0.623–1.475)	0.849
Insulin	1.361 (0.909–2.036)	0.134
BMI		
Normal	Ref.	
Underweight	4.117 (1.199–14.134)	**0.025**
Overweight	1.117 (0.764–1.634)	0.569
Obese	0.66 (0.38–1.145)	0.139
Having diabetes-related complications[€]		
No	Ref.	
Yes	2.174 (1.51–3.131)	**<0.001**
COVID-19 influenced decision about fasting		
No	Ref.	
Yes	1.946 (0.969–3.905)	0.061
Cannot fast even though wanted to fast because of diabetes-related complications		
No	Ref.	
Yes	1.38 (0.708–2.687)	0.344
Praying Taraweeh Prayers during this Ramadan		
No	Ref.	
Yes	1.064 (0.727–1.558)	0.749

Note: AOR = Adjusted odds ratio; CI = Confidence interval; BDT = Bangladeshi Taka.
[€] Diabetes-related complications include retinopathy, nephropathy, neuropathy, and macrovascular difficulties.

every one-unit rise in dietary diversity score [40]. In an another study, a positive relationship was discovered between a reduced odds of depression and balanced diets involving high intake of vegetables, fruits, nuts, legumes, olive oil, and fish, and a limited intake of meat and animal products [57].

In consonant with previous findings [32,58], we found that participants who used insulin for treating diabetes were more likely to experience DD. Previously it was reported by individuals with T2DM that oral treatment was considered less burdensome and insulin most burdensome [59]. Injection pain may generate more unfavorable

Table 4
Association of dietary diversity and fasting with depressive symptoms and diabetes distress.

Variables	Depressive symptoms		Diabetes Distress	
	AOR (95% CI)	p-value	AOR (95% CI)	p-value
Dietary Diversity	0.841 (0.763–0.927)	**0.001**	0.87 (0.791–0.956)	**0.004**
Fasting				
No	1.709 (1.064–2.747)	**0.027**	1.395 (0.87–2.235)	0.167
Yes	Reference		Reference	

psychological effects including DD in insulin-treated patients [58]. Furthermore, living alone was a predominant predictor for both DS and DD in the present study, raising the possibility that lonely people may develop both DS and DD [60].

Our findings also indicate that among individuals with T2DM, those with hypertension and heart disease were more likely to experience both DD and DS. This finding resonates with a previous study which found cardiovascular disease to be predict depression among individuals with T2DM [21]. It is also plausible that having multiple chronic conditions may negatively impact well-being, quality of life, and functioning, and may thus contribute to the development of DS and DD [61]. Also resonating with the current findings, previous research indicates that diabetes-related complications (such as retinopathy, nephropathy, neuropathy, macrovascular difficulties) are linked to DS [10,62] and DD [32]. Arambewela et al. reported that having neuropathy, a diabetes-related complication, was associated with depression among T2DM patients attending a diabetes clinic at a tertiary care hospital [9]. Individuals with diabetes-related complications often experience significant emotional, social, physical, and financial burdens [48] when managing their diabetes, and this might lead to negative psychological consequences like DD and DS. The present study found a significant correlation between DD and DS, which suggests screening for DD in depressed T2DM patients and vice versa may promote more effective interventions.

4.1. Limitations

Our findings should be interpreted in light of study limitations. One significant drawback of our study was the absence of a psychiatric interview to assess DS and DD. We measured DS with the PHQ-9 which is designed for screening and it is not intended to diagnose depression clinically. The convenience sampling approach limits the generalizability of the results and may be prone to selection bias. Biases related to social desirability may also have influenced findings. Also, we could not obtain anthropometric measures as telephones interviews were conducted, and self-reported height and weight measures are subject to reporting biases. However, self-reported height and weight have been found to be generally reliable [63]. The cross-sectional nature of the study does not permit inferring causality to the findings or to consider potential impacts of fasting after Ramadan. Also, we were unable gather data during pre-Ramadan or post-Ramadan period because of the cross-sectional nature of the study. In addition to that, our respondents were Muslim diabetes patients (type 2). In the Bangladeshi context, most of the Muslim Diabetes patients fast because of their strong religious belief despite their physician doesn't recommend so. For that reason, we couldn't compare in groups (fasting group vs. non-fasting groups). Data were collected through a telephone survey during the month of Ramadan. So, we couldn't not collect data on glycemic control of the patients. Moreover, the study did not explore antidepressant usage, which might bias results.

4.2. Public health implications

Despite limitations, this study has public health implications and suggests some important avenues for coping with DD and DS in people with T2DM, paving the way for future work in this area. Although DD and DS both may negatively impact individuals with T2DM, these often go undetected and untreated. Providing psychosocial support to target DD and DS in individuals with T2DM may help improve glycemic control [64,65]. Although a few studies have explored DD and DS during Ramadan fasting in individuals with T2DM [27,29,31], there is no prior study in the Bangladeshi context. Therefore, to the best of our knowledge, this is the first study exploring DD and DS among Bangladeshi individuals with T2DM during Ramadan fasting. Other strengths of the study include a considerable sample size and use of reliable tools to assess DD and DS. The study's findings may be used to advise service providers and policymakers on how to enhance and modify the present guidelines on addressing mental wellbeing in individuals with T2DM.

5. Conclusions

In summary, the study unfolds that the individuals with T2DM who were not fasting experienced more DD and DS during the month of Ramadan. More specifically, those who lived alone, had hypertension or heart disease, and had diabetes-related complications were more likely to experience DD and DS. Our study suggests that fasting may improve DD and DS in individuals with T2DM during Ramadan, although those individuals with less DD and DS may have been more willing or able to fast is another possibility. The findings may help with efforts to address the psychological needs of individuals with T2DM and develop more effective approaches to improve their mental health. Further large-scale research is needed to extend these findings, generate more effective intervention programs and explore their effects on diabetes care during Ramadan fasting.

CRediT authorship contribution statement

Mst. Sadia Sultana: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft, Writing – review & editing, Validation. **Md. Saiful Islam:** Investigation, Formal analysis, Data curation, Writing – original draft, Validation. **Abu Sayeed:** Investigation, Writing – review & editing, Validation. **Marc N. Potenza:** Writing – review & editing, Validation. **Md Tajuddin Sikder:** Supervision, Writing – review & editing, Validation. **Muhammad Aziz Rahman:** Writing – review & editing, Validation. **Kamrun Nahar Koly:** Writing – review & editing, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Atlas D. International diabetes federation. In: IDF Diabetes Atlas. 7th ed. Brussels, Belgium: International Diabetes Federation; 2015.
[2] Guariguata L, Whiting D, Weil C, Unwin N. The International Diabetes Federation diabetes atlas methodology for estimating global and national prevalence of diabetes in adults. Unwin, and c. practice, "The International Diabetes Federation diabetes atlas methodology for estimating global and national prevalence of diabetes in adults," 2011;94(3):322–32.
[3] ICDDRB. (2021, Sep 26, 2021). *Non-communicable diseases*. Available: https://www.icddrb.org/news-and-events/press-corner/media-resources/non-communicable-diseases.
[4] Ali S, Stone MA, Peters JL, Davies MJ, Khunti K. The prevalence of co-morbid depression in adults with Type 2 diabetes: a systematic review and meta-analysis, (in eng). Diabet Med Nov 2006;23(11):1165–73.
[5] Mendenhall E, Norris SA, Shidhaye R, Prabhakaran D. "Depression and type 2 diabetes in low- and middle-income countries: a systematic review," (in eng). Diabetes Res Clin Pract 2014;103(2):276–85.
[6] Bajaj S, Agarwal SK, Varma A, Singh VK. "Association of depression and its relation with complications in newly diagnosed type 2 diabetes," (in eng). Indian J Endocrinol Metab Sep 2012;16(5):759–63.
[7] Joshi S, Dhungana RR, Subba UK. Illness Perception and Depressive Symptoms among Persons with Type 2 Diabetes Mellitus: An Analytical Cross-Sectional Study in Clinical Settings in Nepal. Journal of Diabetes Research 2015;2015:1–9.
[8] Zahid N, Asghar S, Claussen B, Hussain A. "Depression and diabetes in a rural community in Pakistan," (in eng). Diabetes Res Clin Pract 2008;79(1):124–7.
[9] Arambewela MH, Somasundaram NP, Jayasekara H, Kumbukage MP. "Prevalence of Depression and Associated Factors among Patients with Type 2 Diabetes Attending the Diabetic Clinic at a Tertiary Care Hospital in Sri Lanka: A Descriptive Study," (in eng). Psychiatry J 2019;2019:7468363.
[10] Islam SMS, Rawal LB, Niessen LW. "Prevalence of depression and its associated factors in patients with type 2 diabetes: A cross-sectional study in Dhaka, Bangladesh," (in eng). Asian J Psychiatr 2015;17:36–41.
[11] Lustman PJ, Anderson RJ, Freedland KE, de Groot M, Carney RM, Clouse RE. Depression and poor glycemic control: a meta-analytic review of the literature," (in eng). Diabetes Care Jul 2000;23(7):934–42.
[12] Katon WJ, et al. "The association of comorbid depression with mortality in patients with type 2005;28(11):2668–72. 2 diabetes,".
[13] Kokoszka A, Pouwer F, Jodko A, Radzio R, Mućko P, Bieńkowska J, et al. "Serious diabetes-specific emotional problems in patients with type 2 diabetes who have different levels of comorbid depression: a Polish study from the European Depression in Diabetes (EDID) Research Consortium," (in eng). Eur Psychiatry 2009;24(7):425–30.
[14] Fisher L, Mullan JT, Skaff MM, Glasgow RE, Arean P, Hessler D. Predicting diabetes distress in patients with Type 2 diabetes: a longitudinal study. Predicting diabetes distress in patients with type 2 diabetes: a longitudinal study 2009;26(6):622–7.
[15] Diabetes UK. Emotional and Psychological Support and Care in Diabetes, 2010; 2010. Available: https://www.diabetes.org.uk/professionals/position-statements-reports/diagnosis-ongoing-management-monitoring/emotional-and-psychological-support-and-care-in-diabetes [accessed on: 2 February, 2022].
[16] Joensen LE, et al. "Diabetes and COVID-19: psychosocial consequences of the COVID-19 pandemic in people with diabetes in Denmark-what characterizes people with high levels of COVID-19-related worries?," (in eng). Diabet Med Jul 2020;37(7):1146–54.
[17] The World Health Organization. (2021, Aug 17). *COVID-19*. Available: https://cdn.who.int/media/docs/default-source/searo/bangladesh/covid-19-who-bangladesh-situation-reports/who_covid-19-update_63_20210510.pdf?sfvrsn=a2030262_11.
[18] Khan AH, Sultana MS, Hossain S, Hasan MT, Ahmed HU, Sikder MT. The impact of COVID-19 pandemic on mental health & wellbeing among home-quarantined Bangladeshi students: A cross-sectional pilot study, (in eng). J Affect Disord 2020;277:121–8.
[19] Sujan MSH, Tasnim R, Islam MS, Ferdous MZ, Apu MAR, Musfique MM, et al. COVID-19-specific diabetes worries amongst diabetic patients: The role of social support and other co-variates. Primary Care Diabetes 2021;15(5):778–85.
[20] Asghar S, Hussain A, Ali SMK, Khan AKA, Magnusson A. Prevalence of depression and diabetes: a population-based study from rural Bangladesh. Diabetic Med 2007;24(8):872–7.
[21] Roy T, Lloyd CE, Parvin M, Mohiuddin KGB, Rahman M. Prevalence of co-morbid depression in out-patients with type 2 diabetes mellitus in Bangladesh. BMC Psychiatry 2012;12(1):123. 2012/08/22.
[22] Kalra S, et al. National recommendations: Psychosocial management of diabetes in India," (in eng). Indian J Endocrinol Metabol 2013;17(3):376–95.
[23] Jaleel MA, Raza SA, Fathima FN, Jaleel BN. "Ramadan and diabetes: As-Saum (The fasting)," (in eng). Indian J Endocrinol Metab Oct 2011;15(4):268–73.
[24] Jawad F, Kalra S. "Diabetes care in Ramadan: An exemplar of person centered care," (in eng). J Pak Med Assoc May 2015;65(5 Suppl 1):S1–2.
[25] El-Menouar Y. The Five Dimensions of Muslim Religiosity. Methoden, Daten, Analysen, vol. 8, 01/01 2014.
[26] Schielke S. Being good in Ramadan: ambivalence, fragmentation, and the moral self in the lives of young Egyptians. Being good in Ramadan: ambivalence, fragmentation, and the moral self in the lives of young Egyptians 2009;15:S24–40.
[27] Al-Ozairi E, AlAwadhi MM, Al-Ozairi A, Taghadom E, Ismail K. A prospective study of the effect of fasting during the month of Ramadan on depression and diabetes distress in people with type 2 diabetes," (in eng). Diabetes Res Clin Pract Jul 2019;153:145–9.
[28] Shariful Islam SM, Lechner A, Ferrari U, Froeschl G, Niessen LW, Seissler J, et al. Social and economic impact of diabetics in Bangladesh: protocol for a case-control study. BMC Public Health 2013;13(1). https://doi.org/10.1186/1471-2458-13-1217.
[29] Koushali AN, Hajiamini Z, Ebadi A, Bayat N, Khamseh F. "Effect of Ramadan fasting on emotional reactions in nurses," (in eng). Iran J Nurs Midwifery Res May 2013;18(3):232–6.
[30] Lustman PJ, Griffith LS, Clouse RE, Freedland KE, Eisen SA, Rubin EH, et al. "Effects of nortriptyline on depression and glycemic control in diabetes: results of a double-blind. placebo-controlled trial," 1997;59(3):241–50.
[31] Yousuf S, Syed A, Ahmedani MY. To explore the association of Ramadan fasting with symptoms of depression, anxiety, and stress in people with diabetes. Diabetes Res Clin Pract 2021:172. https://doi.org/10.1016/j.diabres.2020.108545.
[32] Islam MR, Karim MR, Habib SH, Yesmin K. Diabetes distress among type 2 diabetic patients. Int J Med Biomed Res 2013;2(2):113–24.
[33] Alberti KG, Zimmet PZ. Definition, diagnosis and classification of diabetes mellitus and its complications. Part 1: diagnosis and classification of diabetes mellitus provisional report of a WHO consultation, (in eng). Diabet Med 1998;15:539–53.
[34] Johnson ST, Al Sayah F, Mathe N, Johnson JA. The relationship of diabetes-related distress and depressive symptoms with physical activity and dietary behaviors in adults with type 2 diabetes: A cross-sectional study. J Diabetes Complications 2016;30(5):967–70.
[35] Islam MR, Sultana MS, Khan AH, Hossain S, Sikder MT, Hasan MT, et al. Fear and depressive symptoms amid COVID-19: A cross-sectional pilot study among adult population in Bangladesh. Heliyon 2021;7(7):e07395. https://doi.org/10.1016/j.heliyon.2021.e07395.
[36] Islam MS, Tasnim R, Sujan MSH, Ferdous MZ, Sikder MT, Masud JHB, et al. Depressive symptoms associated with COVID-19 preventive practice measures, daily activities in home quarantine and suicidal behaviors: Findings from a large-scale online survey in Bangladesh. BMC Psychiatry 2021;21(1). https://doi.org/10.1186/s12888-021-03246-7.
[37] Brzęk A, Ziora KJPP. "A study on the relationship between orthorexia and vegetarianism using the BOT. (Bratman Test for Orthorexia)," 2017;51(6):1133–44.
[38] World Health Organization. Measuring obesityclassification and description of anthropometric data. Report on a WHO consultation of the epidemiology of obesity. Warsaw 21-23 October 1987. Copenhagen: WHO; 1989. Nutrition Unit Document, EUR/ICP/NUT 123.
[39] Kennedy G, Ballard T, Dop M. Guidelines for measuring household and individual dietary diversity. Rome, Italy: FAO, 2011. Available: https://www.fao.org/publications/card/en/c/5aacbe39-068f-513b-b17d-1d92959654ea/.
[40] Poorrezaeian M, Siassi F, Milajerdi A, Qorbani M, Karimi J, Sohrabi-Kabi R, et al. Depression is related to dietary diversity score in women: a cross-sectional study from a developing country. Ann Gen Psychiatry. 2017;16(1). https://doi.org/10.1186/s12991-017-0162-2.
[41] McGuire BE, Morrison TG, Hermanns N, Skovlund S, Eldrup E, Gagliardino J, et al. Short-form measures of diabetes-related emotional distress: the Problem Areas in Diabetes Scale (PAID)-5 and PAID-1. Diabetologia 2010;53(1):66–9.
[42] Polonsky WH, Anderson BJ, Lohrer PA, Welch G, Jacobson AM, Aponte JE, et al. Assessment of Diabetes-Related Distress. Diabetes Care 1995;18(6):754–60.
[43] Snoek FJ, Kersch NYA, Eldrup E, Harman-Boehm I, Hermanns N, Kokoszka A, et al. Monitoring of Individual Needs in Diabetes (MIND): baseline data from the cross-national Diabetes Attitudes, Wishes, and Needs (DAWN) MIND study. Diabetes Care 2011;34(3):601–3.
[44] Kroenke K, Spitzer RL, Williams JBW, Löwe B. The Patient Health Questionnaire Somatic, Anxiety, and Depressive Symptom Scales: a systematic review. Gen Hosp Psychiatry 2010;32(4):345–59.
[45] Chowdhury AN, Ghosh S, Sanyal D. "Bengali adaptation of brief patient health questionnaire for screening depression at primary care," (in eng). J Indian Med Assoc Oct 2004;102(10):544–7.
[46] Sultana MS, Khan AH, Hossain S, Hasan MT. Mental health difficulties in students with suspected COVID-19 symptoms and students without suspected COVID-19

symptoms: A cross-sectional comparative study during the COVID-19 pandemic. Children Youth Serv Rev 2021. https://doi.org/10.1016/j.childyouth.2021.106137.

[47] Sultana MS, Khan AH, Hossain S, Ansar A, Sikder MT, Hasan MT. Prevalence and predictors of post-traumatic stress symptoms and depressive symptoms among Bangladeshi students during COVID-19 mandatory home confinement: A nationwide cross-sectional survey. Children and Youth Services Review 2021/03/01/ 2021.;122:105880.

[48] Kalra S, Jena BiranchiNarayan, Yeravdekar R. Emotional and psychological needs of people with diabetes. Indian J Endocr Metab 2018;22(5):696. https://doi.org/10.4103/ijem.IJEM_579_17.

[49] Fisher L, Glasgow RE, Mullan JT, Skaff MM, Polonsky WH. Development of a Brief Diabetes Distress Screening Instrument. The Annals of Family Medicine 2008;6(3):246–52.

[50] Islam SMS, Rawal LB, Niessen LW. Prevalence of depression and its associated factors in patients with type 2 diabetes: A cross-sectional study in Dhaka, Bangladesh. Asian Journal of Psychiatry 2015;17:36–41.

[51] Leone T, Coast E, Narayanan S, de Graft Aikins A. Diabetes and depression comorbidity and socio-economic status in low and middle income countries (LMICs): a mapping of the evidence. and health, "Diabetes and depression comorbidity and socio-economic status in low and middle income countries (LMICs): a mapping of the evidence," 2012;8(1):39. https://doi.org/10.1186/1744-8603-8-39.

[52] Prathikanti S, Rivera R, Cochran A, Tungol JG, Fayazmanesh N, Weinmann EJPo. Treating major depression with yoga: A prospective, randomized, controlled pilot trial. PLoS One 2017;12(3):e0173869. https://doi.org/10.1371/journal.pone.0173869.

[53] Hallahmi BB, Argyle M. The Psychology of Religious Behaviour, Belief and Experience. London: Routledge; 1998.

[54] Kawachi I, Berkman LF. Social ties and mental health. J Urban Health 2001 Sep;78(3):458-67. doi: 10.1093/jurban/78.3.458. PMID: 11564849; PMCID: PMC3455910.

[55] de Wit LM, van Straten A, van Herten M, Penninx BW, Cuijpers P. Depression and body mass index, a u-shaped association. BMC Public Health 2009 Jan 13;9:14. doi: 10.1186/1471-2458-9-14. PMID: 19144098; PMCID: PMC2631467.

[56] T. Sal-Sabil, A. Islam, and S. S. J. J. o. D. Islam, "Risk factors for type 2 diabetes in Bangladesh: a systematic," vol. 2, p. 5, 2016.

[57] Rashidkhani B, Pourghassem Gargari B, Ranjbar F, Zareiy S, Kargarnovin Z. Dietary patterns and anthropometric indices among Iranian women with major depressive disorder. Dietary patterns and anthropometric indices among Iranian women with major depressive disorder 2013;210(1):115–20.

[58] Rahman M, Rahman MA, Flora MS, Rakibuz-Zaman M. Depression and associated factors in diabetic patients attending an urban hospital of Bangladesh. Int J Collaborat Res Internal Med Public Health 2011;3:65–76.

[59] Vijan S, Hayward RA, Ronis DL, Hofer TP. BRIEF REPORT: The burden of diabetes therapy: Implications for the design of effective patient-centered treatment regimens. J Gen Intern Med 2005;20(5):479–82.

[60] Richard A, Rohrmann S, Vandeleur CL, Schmid M, Barth J, Eichholzer M. Loneliness is adversely associated with physical and mental health and lifestyle factors: Results from a Swiss national survey, (in eng). PLoS One 2017;12(7):e0181442. https://doi.org/10.1371/journal.pone.0181442.

[61] de Groot M, Anderson R, Freedland KE, Clouse RE, Lustman PJ. Association of Depression and Diabetes Complications: A Meta-Analysis: Psychosom Med 2001;63(4):619–30.

[62] Anderson RJ, Freedland KE, Clouse RE, Lustman PJ. The prevalence of comorbid depression in adults with diabetes: a meta-analysis. Diabetes Care. 2001 Jun;24(6):1069-78. doi: 10.2337/diacare.24.6.1069. PMID: 11375373.

[63] Wada K, Tamakoshi K, Tsunekawa T, Otsuka R, Zhang H, Murata C, et al. Validity of self-reported height and weight in a Japanese workplace population. Int J Obes (Lond). 2005 Sep;29(9):1093-9. doi: 10.1038/sj.ijo.0803012. PMID: 15925952.

[64] Ismail K, Winkley K, Rabe-Hesketh S. "Systematic review and meta-analysis of randomised controlled trials of psychological interventions to improve glycaemic control in patients with type 2 diabetes," (in eng). Lancet 2004;363(9421):1589–97.

[65] Peyrot M, Rubin RR. "Behavioral and psychosocial interventions in diabetes: a conceptual review," (in eng). Diabetes Care Oct 2007;30(10):2433–40.